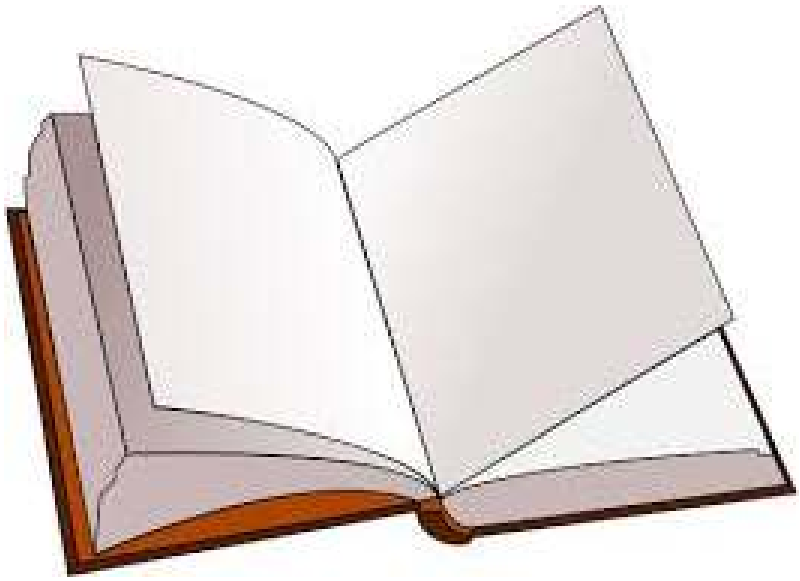
The Reddit Study Guide



Good students: How do you go about getting good grades? [Serious] (self.AskReddit)
submitted 1 day ago * by irollon

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18 July 14, Friday
Asked by irollon
Collated by Salticido
Exported to Word by SailboatoMD

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Salticido 3625 points 21 hours ago*x8

HOW TO STUDY BASED ON HOW MEMORY WORKS

Memory works (to put it simply) in 3 stages: **attention**, **encoding** (storing/associating with other info), and **retrieval**(remembering)

To optimize the final stage, you have to optimize the first two stages. This means you have to pay attention to the material, and you have to encode it well. (Which I'll explain below.) Additionally, if you repeat the process, you reinforce it. By retrieving something, you start to pay attention to it again, and then you are able to re-encode it better than before.

To optimize encoding, remember GOAT ME.

- G is generate and test. i.e., quiz yourself, or otherwise come up with the answers on your own without just reading them. Even if you get it wrong, it helps more than if you just read the answer off the bat, because you're forcing yourself to think more about it (why was it wrong?). Test yourself in a way that will resemble what you'll actually have to do during the real test. (e.g., if you have to write essays on the test, instead of just writing and memorizing bullet points, actually write an essay multiple times without cheating, review it, and repeat until you can write it without forgetting any important points.) Other effective ways of testing yourself are teaching the material to someone else and talking about it out loud to yourself.
- O is organize. This reduces the load on your brain and helps create reminders just by coloring, position, or associations with nearby material. For instance, a time line helps remember that event A came before event B in history, not necessarily because you memorized the dates but because you organized the info so that event A was written earlier and you happen to remember that it was written earlier. The position of the information becomes meaningful. You can organize with outlines, pictures, color coding, related material, etc. My use of "GOAT ME" can be thought of as organization. Another fun example (chunking) is as follows. Which of these seems easier to memorize: "CIAFBIKGBCNNUSABBCUK" or "CIA FBI KGB CNN USA BBC UK"?
- A is for avoid illusions of learning. There are two kinds of memory: familiarity/recognition and recall. Recall is what you want. That's where you can remember the information on your own, as you might be expected to do on a test. Recognition is

where you can't think of it on your own but if you see it you suddenly remember it. That's not good. You won't necessarily see it on your test, so you won't get a blatant reminder of it. Avoid study methods that rely on recognition. Similarly, a major problem with rereading material is "fluency". The more you read it, the easier reading it becomes, and when it feels easier to read, you assume you have learned it. You have not. You've just become more skilled at reading it. Don't bother highlighting your textbook in the first go either. You feel like you're picking out the important parts of the chapter but you can't know what's really important until you've read the whole thing. And then all you're gonna do anyway is go back and reread all the highlights, and as we've established, rereading is useless. If instead you actually organize the highlights and quiz yourself on them, highlighting may be useful. For a similar reason, rewriting information is also not very helpful unless you use it as a method of quizzing.

- T is take breaks. This is HUGE. If nothing else, walk away with just this tip. Your memory works best if you study in frequent, short sessions rather than one long cram session. You don't give your brain a chance to store the earlier info you studied, so it just slips out of your mind, and you'll have wasted your time studying it. So study for awhile, go do something else for a bit, and come back to it, and repeat. One of my students said she taped information in front of her toilet so whenever she went to pee or something she could study for just a couple minutes. It sounds strange but it's actually a great idea (I'd advise, in line with G and A that you tape questions in front of the toilet and tape answers elsewhere so you can quiz yourself.) Another important part of this is that you need to sleep to keep that info in your head. Even if you take regular breaks, an all nighter will do more harm than good. Your memories are stored more permanently after sleep. This is just how the brain works. You can even try to work naps into your study sessions. It's a break + sleep! [EDIT: I do not know how long breaks SHOULD be, but I believe this varies from person to person. Just try to study over the course of days instead of hours.]
- M is match learning and testing conditions. This is based off the principle of encoding specificity, which states that, if you want to optimize memory, then the conditions surrounding encoding (e.g., where you are when you study, how tired you are when you study, etc.) should be the same as those surrounding retrieval (e.g., where you are when you're tested,

how tired you are when you're tested, etc.). This is because the conditions themselves serve as reminders. (Have you ever walked into the kitchen for something, forgotten why you were there, and as soon as you return to the other room you suddenly remember why you went to the kitchen?) This includes your environment and your physiology, serving as reminders. Think about noise level, size of room, lighting, types of furniture, mood, intoxication, sitting position, and even the way you work with the material (remember G and A). Studies show that learning while drunk is best remembered while drunk again. Learning after exercising, also best remembered after exercising. **The alternative to this is that you should study under MANY different conditions.** This way, the information comes easily to you regardless of your surrounding conditions. Otherwise, the information will unfortunately be associated with the specific circumstances you studied under and will be difficult to remember in any other situation. If you want to remember this stuff outside of being tested in class, STUDY UNDER MANY CONDITIONS. Study in a noisy place AND a quiet place, with and without coffee, etc.

- E is elaborate. Think deeply about the material and make other associations with it. At the most extreme, this can mean truly understanding the concept, why it works, how it relates to other concepts, and how it's applied. But on a simpler level, it can be the following: Does it remind you of something else? Can you make a song out of it? Can you visually imagine it? How does it apply to you or your life? Instead of taking the material at face value, do something with it. The reason this is important is because of reminders. Memory works by having a network of associations. One thing reminds you of another. If you've thought deeply about it, you've probably associated it with something else in memory, which can then serve as a reminder. You can think, "Oh yeah, this is the term that inspired me to draw that silly stick figure to represent it. And I remember what the drawing looked like so now I remember what the term means." Additionally, the *quality*of the memory will be better if you have elaborated on it. Elaboration allows for a lot of creativity and individuality among studiers. Choose whichever method of elaboration works for you. Maybe you enjoy making up songs, drawing doodles, creating stories, visually imagining it, relating it to yourself, or just pondering about it. If you're studying history, you might try to think about it visually, imagine what people would have said or

looked like, watch them in your head doing their historical stuff, or maybe you'd like to draw a quick doodly comic about a particular event, or maybe you wanna think about why this even was significant, or how it relates to another historical event.

If I had to summarize this in fewer points:

- Keep similar conditions during studying and testing. This includes environmental surroundings, mental and physiological state, the way you think about the material, and so on. *But if you want to remember this outside of class, study in a VARIETY of conditions, so that you don't associate the material with any particular condition.*
- Study briefly and frequently, and sleep.

But one other good point I would add is this:

- Take notes BEFORE class *if possible*, and add to them whenever necessary. Do this by reading the textbook chapters ahead of time (and take notes; refer to your syllabus to find out which chapter is next, if applicable) or see if your teacher posts Powerpoints online ahead of time. This way, you're not just frantically writing notes in class and you'll actually be able to more fully pay attention to what the teacher is saying (remember: attention is the first step of the memory process!). You may think you can pay attention to the professor as you're writing, but you are actually *dividing* your attention and hurting your memory.

EDIT: Whoa, thanks for all the comments, the gold, and the upcoming pizza(s)! I'm trying to get to those who've asked questions, but my inbox has exploded, so sorry if I take awhile! I will try to edit as people offer other good points, but I'm already super close to the character limit! Trying to cut it down now.

gotstonoe 135 points 12 hours ago*

- G is generate and test. Quiz yourself and teach others.
- O is organize. You can organize with outlines, pictures, color coding, related material, etc.
- A is for avoid illusions of learning. Avoid study methods that rely on recognition. If you can't remember it at any given time you don't know it. Learn not memorize or familiarize.
- T is for Take Breaks. Make sure you take breaks and sleep. You can't remember large chunks on information in one sitting. Stand up and come back to it.

- M is match learning and testing conditions. Learn in similar conditions as when you will take the test. *my tip is to chew gum when studyign a certain subject and chew the same flavor of gum during the test. It will help you remember*
- E is elaborate. Think deeply about the material and make other associations with it.

tl;dr Sleep, take breaks, take good notes, learn the material not memorize, and keep conditions the same during studying and testing

edit: personal tip. Keep hydrated and exercise. It helps your memory

lshdevanarchist 8 points 11 hours ago

PhD in Psychology, somewhat expert on learning and part-time college professor here. His advice is great, but I thought it might help to sort out one point. His suggestion to make your study environment as similiar as possible to the test environment is called state dependent learning. This is the best thing to do if all you want is to remember for this class and not remember it after that. If you need to remember it for a long time, like using it on the job or taking a comprehensive exam, you will want to study in several different kinds of environments.

Here is a list of study tips I give my students.

Most of them are from the article -- DUNLOSKY, JOHN; RAWSON, KATHERINE A.; MARSH, ELIZABETH J.;NATHAN, MITCHELL J.; WILLINGHAM, DANIEL T. What Works, What Doesn't. Scientific American Mind. Sep/Oct2013, Vol. 24 Issue 4, p47-53. Sorry I didn't make this a link but I am new here and don't have time to figure that out right now.

Most Effective Techniques

- Practice testing — any form that allows you to test yourself, including using actual or virtual flashcards, doing problems or questions at the end of textbook chapters, or taking practice tests.
- Distributed practice — studying material over a number of relatively short sessions. The best way is to study a section, sleep, then test yourself on that section

Moderately effective

- Elaborative interrogation — use “why” questions to make connections between new and old material.
- Self-explanation — provide your own explanations for problems while learning material
- Interleaved practice — mixing different kinds of problems or material in one study session

Least effective

- Highlighting and underlining textbooks and other materials
- Rereading
- Summarization
- Keyword mnemonics — the use of keywords and mnemonics to help remind students of course material
- Imagery use for text learning — creating mental images to remind students of material

hawkian 4 points 11 hours ago

Might be a bit late for anyone to see this, but here goes anyway!

These are awesome tips for studying, and the best thing about them is they do really hold true regardless of learning style.

However, in response to the OP's question I'd add that a lot of my success in getting good grades throughout high school and college was **getting a feel for each individual teacher** and what they value most, what they consider acceptable effort, where they'll notice that you went above and beyond the average, and what they tend to de-emphasize in an assignment. Any time you're using your own words to give a response- even when asked a basic factual question with "right" and "wrong" answers- teachers have a lot of leeway and discretion in *how* they evaluate a given response and what kind of things they'll consider worthwhile for credit toward a final grade.

Some teachers may prefer an avalanche of information in response to a question with any degree of ambiguity, so that you cover all possible bases when giving an answer. Others may only be looking for their own personally-tailored version of the correct answer; for these classes it's essential to pay attention not just to the information but *how the teacher phrases* this information, so that you can recognize it or reproduce it on a test. Some teachers may love it when you put in answer in broader context, giving a little more information than was asked for in order to demonstrate mastery, while others do not value this at all and you'd be much better off spending your time otherwise.

This may seem like I'm advocating a sort of "gaming the system" or manipulating your teachers rather than learning the material, but I really believe this is both practical and relatively benign advice. Teachers are individuals and just don't all care equally about the same things. Mastering the topic in question will be the difference between passing and failing 100% of the time, but knowing my audience was often the difference between a B+ and an A.

It's fantastic to have a set of guidelines for how you can approach learning in any class; a sort of baseline to apply before you have any idea about the nuances of your instructor. But after a few assignments and quizzes, try to get a feel for what it seems like they value and emphasize the most and the least, and then play to that for the rest of the course.

To summarize by way of analogy, let's pretend that your class is a game of poker. /u/Salticido's post is a magnificent primer on basic strategy that everyone should bear in mind before sitting down at the table. My advice, on the other hand, would be **play the opponent, not just the cards.**

The_White_Baron 4 points 11 hours ago*

I just want to say that this is a great technique for humanities and reading intensive courses. It simply won't work for courses in physics or math though. You've listed a great technique on how to take tests, which *will* help people become great students, but it's not a good way to *permanently retain the material*. You need to keep on studying it **after you've graduated** and completed your course - something most will not do. And why do we go to school? If you have a PhD in sights, it's important you internalize the material. Many TAs who got through with great grades become a TA their first year in grad school and need to *relearn the material* because they forgot it.

Many brilliant students who passed with 3.5+ GPAs in fields step away for a few decades and completely forget the material. The rate of memory loss is actually pretty strong. After a few months you've lost over 80% of the material, which is why transferring it to long term memory is essential. This isn't always always feasible.

Your techniques are also all *different* techniques. They're not intended to work in tandem. They **may** work in tandem, but there is absolutely no requisite that they do.

Don't get me wrong, I agree with everything you say (as I also took a course in studying techniques and it was pretty much memory based). I'm just warning anyone who may read my comment that your comment isn't absolute, and it is a bit scattered. For instance, encoding probably the de facto method for storing materials into your working memory (not exactly short term memory, but for most uses, it is). The real effort is transferring this knowledge to your long term memory, and that's where the techniques come in. The most effective is elaboration.

You need to be *very* careful to suggest that *different material requires different techniques*.

You've got the basic outline of the course down pat, and you've even memorized it and explained it using the techniques you learned, but you missed some of the nuances (which get missed by these techniques).

Though, one thing I will absolutely support and not question is the efficacy of breaks; breaks help with everything. I*always* tell people that an all nighter is the worst thing you can do. Get a good night's rest instead

bestkind0fcorrect 2 points 11 hours ago

This is great information. One of the absolute best ways to do A, T, and E is with group studying. When you study by yourself, you are likely to just reread your notes or the book, but when you study with others, you end up describing the material to others, skipping around,

and having others explain things to you. This also helps with E; you end up teaching something you understand to someone who doesn't, so you have to figure out how to describe it in your own words or come up with analogies/examples. Group studying also usually ends up in the occasional off-topic discussion or coffee break to give you a break from the material. But, because the whole purpose of the meeting is to study, these breaks are usually pretty short, rather than the 2 hour break you might end up taking on your own.

optimismizer – Teaching/Learning Mechanics (Book)

StilesAjax 3 points 13 hours ago

Here are a couple of hint/hacks for Salticido's "Elaborate":

1. Prefer associations that have big feelings. It's easier to remember something funny, arousing, scary, bizarre, etc. I learn better when I associate stuff with sex, murder, totally awesome daydreams. That works for the songs, doodles, etc., too.
2. Repetition can be really helpful for learning. So I try to associate my elaborations with objects or habits.

Here's an example of how these two hacks might work.

I need to learn the five emperors in Chinese mythology.

1. Shaohao. Makes me think of what a messed up dog might say. (Picture a really funny-looking dog, thick glasses, slobbery, maybe the sides of his jaw don't quite work, or he's missing half of his tongue...)
2. Zhuanxu. Wikipedia says he's the grandson of the Huang Emperor. I don't know this word sounds like, but I recognize the 'uan', which makes me think of Juan. I think of a person I know (or an athlete, etc.) called Juan, and imagine him so well hung (Huang) that when he walks, his junk drags on the ground. I'll know that 'Juan' isn't the right answer, and I'm pretty sure I can remember that it starts with Zh and ends in 'xu'.

Then I put those two together. There's a messed-up dog saying "Shaohao" (trying to say bow-wow), who's very interested in Juan's giant crotch. Juan is trying to get away. But that crotch weighs him down, so he's not terribly mobile.

1. Emperor Ku. Ku makes me think of Kung Fu. I imagine Juan/Zhuanxu trying to do Kung Fu on the dog, but not being able to jump. ---
   (aside: ku looks like xu, if a spear sliced its left half off. which ever sticks first, the 'ends in xu' or the 'ku', I can use to remember the other.)
2. Yao. While Juan/Zhuanxu is trying to jump, the dog bites off his third leg. He yells, "Yow!!!"
3. Shun. Juan/Zhuanxu and the dog are both shunned by society, and they wind up having to live together in a shared giant dog house.

So far, I've made the elaboration memorable. Now I need to tie it to an object or habit, so I can use repetition to make it cement.

So... I tie the story to something I do all the time, or an object I see fairly frequently:

I imagine that when I swipe to unlock my phone, I'm slicing off Zhuanxu/Juan's package, a la Fruit Ninja... or that it makes the big messed up dog come out of the phone, and his slobber might get my home button wet.

And now when I unlock the phone, I remember to take a few seconds to review the story, and the 5 emperors: Shaohao, Zhuanxu, Ku, Yao, and Shun.

Tho76 1 point 14 hours ago

> “This way, you're not just frantically writing notes in class and you'll actually be able to more fully pay attention to what the teacher is saying”

This is how I do well in class. To me, it is a higher priority to hear what the teacher is saying. The teacher is the one making the test, they will often type their answers as the exact words they say in class, which makes them seem more familiar. Raw facts are easy to look up.

I'm not going to lie, I skimmed the wall of text. But I don't see you mention this anywhere - there are three main types of learners: Audio, Visual, and Kinetic. Audio learners learn best by hearing, Visual by seeing, Kinetic by doing. Everyone is different, and if you know what type you are it can really help your studying. I am a Kinetic, so when learning, for example, Equilibrium in Chem I would physically take some salt and some pepper and try to understand it.

The other thing I do is try to understand the concepts (for math mainly). It's sort of like looking at it from a different perspective. I did better than others in my Chem/Physics class because people would just try to memorize facts/formulas, but I would try to understand what they meant, which I could use to help me know the formula

Jstbcool 1 point 14 hours ago

Overall I really like this comment, but I would change the part about studying 20 minutes at a time and then doing something else. That is not going to help your memory and frequently switching between tasks can reduce your attentional resources much quicker. Study for an hour or 2 (as long as you dont feel bored and you're engaged in the material) and then take a break or do something fun.

I would also add avoid having the TV on or listening to music with vocals as you will have trouble effectively encoding the words you're reading. Your brain is getting 2 sources of input and its going to have to choose which one to encode.

Source: Psychologist who studies memory.

Roez 4 points 13 hours ago*

A great post. This information was available over 25 years ago when I was in college, and it works.

I was a straight A student the majority of my undergraduate, graduate and professional degrees. I rarely studied during finals, except to just review my notes to reinforce I knew the material enough I didn't need worry. Essentially, following the principles above this is what I did:

- If there's reading, do it before class, and then ask yourself what you read. Some classes rely on reading more than others. For the most part, it's a first step in starting to learn material and be able to recall it earlier. Time is your friend.
- Review your class notes within 30 to 60 minutes after each class. It helps reinforce the material you learned and increases memory by an incredible percent. I found I actually could discover mistakes, or points I needed to clarify, within my own notes because my recollection was still fresh.
- Change subjects and take breaks after 60 to 90 minute sessions. Try to spread your studying out throughout the day.
- Make sure to not study one subject one day and then ignore it for two or three days, you will have wasted a lot of your time. Even just fifteen minutes on each subject every day or two can help you recall a lot of what you previously learned, and help reinforce its retention.
- Test your own understanding. This is the recall above, as well as the generate and test. Examples: Rewrite your own notes, try to put the notes in your own words that are accurate. Work through problems your professor didn't assign. Find people who took the class before and see if you can see their old tests, and then practice taking those tests.
- Keep a little diary or log book, and mark the time you spend studying each day and total up it up each week. It helps you keep a schedule and not ignore one subject. It also allows you to positively reinforce success, and not let you convenience yourself you studied hard when you didn't.

As an aside, a good student will follow good study habits after school. At work they'll likely find they are able to retain and collect new information quickly, and know how to prepare projects, remember speeches, be more efficient, etc. These habits have life long benefits.

rohmer95 2 points 13 hours ago*

I'd just like to add that students should certainly study different things under different conditions, other things equal, instead of focusing on being in one spot.

For a helpful study guide to studying (isn't that meta?) see this article, which Harvard's undergrad intro to psychology class hands out along with the class syllabus:

http://www.nytimes.com/2010/09/07/health/views/07mind.html

"The brain makes subtle associations between what it is studying and the background sensations it has at the time, the authors say, regardless of whether those perceptions are conscious. It colors the terms of the Versailles Treaty with the wasted fluorescent glow of the dorm study room, say; or the elements of the Marshall Plan with the jade-curtain shade of the willow tree in the backyard."

In general though, I think these things are fairly small points - most of the people around me who got straight A's all their lives do the 'cram in one day' approach and usually do fine. I think it's largely about motivation and about seeking out extra help (via office hours, usually). I think the people who tend to do the BEST, though (i.e. an A or high A- average), work really hard all semester, consistently meet with Professors, ask questions about things they are 99% (but not 100%) sure of just to get to 100%, etc.

Forget What You Know About Good Study Habits

By BENEDICT CAREY

Every September, millions of parents try a kind of psychological witchcraft, to transform their summer-glazed campers into fall students, their video-bugs into bookworms. Advice is cheap and all too familiar: Clear a quiet work space. Stick to a homework schedule. Set goals. Set boundaries. Do not bribe (except in emergencies).

And check out the classroom. Does Junior's learning style match the new teacher's approach? Or the school's philosophy? Maybe the child isn't "a good fit" for the school.

Such theories have developed in part because of sketchy education research that doesn't offer clear guidance. Student traits and teaching styles surely interact; so do personalities and at-home rules. The trouble is, no one can predict how.

Yet there are effective approaches to learning, at least for those who are motivated. In recent years, cognitive scientists have shown that a few simple techniques can reliably improve what matters most: how much a student learns from studying.

The findings can help anyone, from a fourth grader doing long division to a retiree taking on a new language. But they directly contradict much of the common wisdom about good study habits, and they have not caught on.

For instance, instead of sticking to one study location, simply alternating the room where a person studies improves retention. So does studying distinct but related skills or concepts in one sitting, rather than focusing intensely on a single thing.

"We have known these principles for some time, and it's intriguing that schools don't pick them up, or that people don't learn them by trial and error," said Robert A. Bjork, a psychologist at the University of California, Los Angeles. "Instead, we walk around with all sorts of unexamined beliefs about what works that are mistaken."

Take the notion that children have specific learning styles, that some are "visual learners" and others are auditory; some are "left-brain" students, others "right-brain." In a recent review of the relevant research, published in the journal Psychological Science in the Public Interest, a team of psychologists found almost zero support for such ideas. "The contrast between the enormous popularity of the learning-styles approach within education and the lack of credible evidence for its utility is, in our opinion, striking and disturbing," the researchers concluded.

Ditto for teaching styles, researchers say. Some excellent instructors caper in front of the blackboard like summer-theater Falstaffs; others are reserved to the point of shyness. "We have yet to identify the common threads between teachers who create a constructive learning atmosphere," said Daniel T. Willingham, a psychologist at the University of Virginia and author of the book "Why Don't Students Like School?"

But individual learning is another matter, and psychologists have discovered that some of the most hallowed advice on study habits is flat wrong. For instance, many study skills courses insist that students find a specific place, a study room or a quiet corner of the library, to take their work. The research finds just the opposite. In one classic 1978 experiment, psychologists found that college students who studied a list of 40 vocabulary words in two different rooms — one windowless and cluttered,

the other modern, with a view on a courtyard — did far better on a test than students who studied the words twice, in the same room. Later studies have confirmed the finding, for a variety of topics.

The brain makes subtle associations between what it is studying and the background sensations it has at the time, the authors say, regardless of whether those perceptions are conscious. It colors the terms of the Versailles Treaty with the wasted fluorescent glow of the dorm study room, say; or the elements of the Marshall Plan with the jade-curtain shade of the willow tree in the backyard. Forcing the brain to make multiple associations with the same material may, in effect, give that information more neural scaffolding.

"What we think is happening here is that, when the outside context is varied, the information is enriched, and this slows down forgetting," said Dr. Bjork, the senior author of the two-room experiment.

Varying the type of material studied in a single sitting — alternating, for example, among vocabulary, reading and speaking in a new language — seems to leave a deeper impression on the brain than does concentrating on just one skill at a time. Musicians have known this for years, and their practice sessions often include a mix of scales, musical pieces and rhythmic work. Many athletes, too, routinely mix their workouts with strength, speed and skill drills.

The advantages of this approach to studying can be striking, in some topic areas. In a study recently posted online by the journal Applied Cognitive Psychology, Doug Rohrer and Kelli Taylor of the University of South Florida taught a group of fourth graders four equations, each to calculate a different dimension of a prism. Half of the children learned by studying repeated examples of one equation, say, calculating the number of prism faces when given the number of sides at the base, then moving on to the next type of calculation, studying repeated examples of that. The other half studied mixed problem sets, which included examples of all four types of calculations grouped together. Both groups solved sample problems along the way, as they studied.

A day later, the researchers gave all of the students a test on the material, presenting new problems of the same type. The children who had studied mixed sets did twice as well as the others, outscoring them 77 percent to 38 percent. The researchers have found the same in experiments involving adults and younger children.

¶"When students see a list of problems, all of the same kind, they know the strategy to use before they even read the problem," said Dr. Rohrer. "That's like riding a bike with training wheels." With mixed practice, he added, "each problem is different from the last one, which means kids must learn how to choose the appropriate procedure — just like they had to do on the test."

¶These findings extend well beyond math, even to aesthetic intuitive learning. In an experiment published last month in the journal Psychology and Aging, researchers found that college students and adults of retirement age were better able to distinguish the painting styles of 12 unfamiliar artists after viewing mixed collections (assortments, including works from all 12) than after viewing a dozen works from one artist, all together, then moving on to the next painter.

¶The finding undermines the common assumption that intensive immersion is the best way to really master a particular genre, or type of creative work, said Nate Kornell, a psychologist at Williams College and the lead author of the study. "What seems to be happening in this case is that the brain is picking up deeper patterns when seeing assortments of paintings; it's picking up what's similar and what's different about them," often subconsciously.

¶Cognitive scientists do not deny that honest-to-goodness cramming can lead to a better grade on a given exam. But hurriedly jam-packing a brain is akin to speed-packing a cheap suitcase, as most students quickly learn — it holds its new load for a while, then most everything falls out.

¶"With many students, it's not like they can't remember the material" when they move to a more advanced class, said Henry L. Roediger III, a psychologist at Washington University in St. Louis. "It's like they've never seen it before."

¶When the neural suitcase is packed carefully and gradually, it holds its contents for far, far longer. An hour of study tonight, an hour on the weekend, another session a week from now: such so-called spacing improves later recall, without requiring students to put in more overall study effort or pay more attention, dozens of studies have found.

¶No one knows for sure why. It may be that the brain, when it revisits material at a later time, has to relearn some of what it has absorbed before adding new stuff — and that that process is itself self-reinforcing.

“The idea is that forgetting is the friend of learning,” said Dr. Kornell. “When you forget something, it allows you to relearn, and do so effectively, the next time you see it.”

That’s one reason cognitive scientists see testing itself — or practice tests and quizzes — as a powerful tool of learning, rather than merely assessment. The process of retrieving an idea is not like pulling a book from a shelf; it seems to fundamentally alter the way the information is subsequently stored, making it far more accessible in the future.

Dr. Roediger uses the analogy of the Heisenberg uncertainty principle in physics, which holds that the act of measuring a property of a particle (position, for example) reduces the accuracy with which you can know another property (momentum, for example): “Testing not only measures knowledge but changes it,” he says — and, happily, in the direction of more certainty, not less.

In one of his own experiments, Dr. Roediger and Jeffrey Karpicke, who is now at Purdue University, had college students study science passages from a reading comprehension test, in short study periods. When students studied the same material twice, in back-to-back sessions, they did very well on a test given immediately afterward, then began to forget the material.

But if they studied the passage just once and did a practice test in the second session, they did very well on one test two days later, and another given a week later.

“Testing has such bad connotation; people think of standardized testing or teaching to the test,” Dr. Roediger said. “Maybe we need to call it something else, but this is one of the most powerful learning tools we have.”

Of course, one reason the thought of testing tightens people’s stomachs is that tests are so often hard. Paradoxically, it is just this difficulty that makes them such effective study tools, research suggests. The harder it is to remember something, the harder it is to later forget. This effect, which researchers call “desirable difficulty,” is evident in daily life. The name of the actor who played Linc in “The Mod Squad”? Francie’s brother in “A Tree Grows in Brooklyn”? The name of the co-discoverer, with Newton, of calculus?

The more mental sweat it takes to dig it out, the more securely it will be subsequently anchored.

None of which is to suggest that these techniques — alternating study environments, mixing content, spacing study sessions, self-testing or all the above — will turn a grade-A slacker into a grade-A student. Motivation matters. So do impressing friends, making the hockey team and finding the nerve to text the cute student in social studies.

"In lab experiments, you're able to control for all factors except the one you're studying," said Dr. Willingham. "Not true in the classroom, in real life. All of these things are interacting at the same time."

But at the very least, the cognitive techniques give parents and students, young and old, something many did not have before: a study plan based on evidence, not schoolyard folk wisdom, or empty theorizing.

This article has been revised to reflect the following correction:

Correction: September 8, 2010.

An article on Tuesday about the effectiveness of various study habits described incorrectly the Heisenberg uncertainty principle in physics. The principle holds that the act of measuring one property of a particle (position, for example) reduces the accuracy with which you can know another property (momentum, for example) — not that the act of measuring a property of the particle alters that property

irollon[S] 286 points 21 hours ago

What else have you learned about memory?? :)

[–]**Salticido** 567 points 20 hours ago

The way you're socialized has a big impact on what you remember, or at least what you report. Parents elaborate on memories with little girls more than little boys, so girls end up with more elaborate memories. They also tend to remember more about feelings, where as boys remember more about autonomous activities. There are also cultural differences in the age of the earliest memory. (First memory is around 3.8 years old in US and 5.4 years old in China, probably because mothers are more elaborative in the US than in China when talking to their kids about what they did).

False memories are fairly common, and it's possible to create them, though some people are more susceptible than others.

Memory is not like a perfect tape recorder that you can just play back. It's a work in progress. Every time you recall something, you store it in your memory differently than it was before. This is actually great because if your memory was wrong, you can update it.

Having a super good memory is not necessarily a good thing. Check out this excerpt from the textbook: "AJ remembers every single day of her life since her teens in extraordinary detail. Mention any date … and she finds herself … reliving events and feelings as though they happened yesterday. She can tell you what day of the week it was, events that took place on all surrounding days, and intricate details about her thoughts, feelings, and public events … AJ reports that these memories are vivid ... and full of emotion. Her remembering feels automatic, and not under conscious control … When unpleasant things happen, AJ wishes she could forget, and the constant bombardment by reminders is distracting and sometimes troubling."

There's also stuff about how memory and your sense of self influence each other. Stuff about how you are better at remembering the appearance of people withing your own group (same age, race, etc.). And a bunch more. It was seriously a whole class. Haha.

splooshh 476 points 18 hours ago*

- **DO IT.** Do it now, not later, not tomorrow. Oh, there is a new episode of your favourite show that you HAVE to watch although there is work to be done? Guess what, I don't care. If there is stuff to be done, then do it. If you don't do it, someone else will and these people will be a lot more successful than you in the long run. The things you do now that other people don't do will enable you to do things that other people won't be able to do later in life. Get that Nike mantra: JUST DO IT!
- Pay attention in class and write stuff down. There are only a few people out there who can remember everything that is important without writing it down and I highly doubt that you are one of them. Once you are at home, you eat and do what? Go to the computer and browse reddit all day or play games? Nope, you review the things you have written down and try to explain them to yourself. You aren't able to do that? Next lesson go to your teacher and ask him if he could explain it again. Not next week or one day before you have an exam, you ask him when the next lesson starts. PARTICIPATE IN CLASS!
- Do your homework and understand what you are doing. So many people do things and have no clue why they do it or what they can do with the things they've just learned. Try to understand things and explain them to people. If you can explain something to someone, you understand it.
- FOCUS! Many people really want to study but procrastinate all the time. If you can't focus in your room, go into a library or somewhere else where you are not distracted.
- Nutrition and workout - A healthy lifestyle helps you to concentrate, makes you feel better and is good for your body.
- Take breaks, enjoy your life (socialise) and relax. Try to find a hobby where you can learn something that will help you in the long run. Read.

Something that may help you is the 3 Seconds Rule: http://getbusylivingblog.com/how-to-start-anything-and-get-unstuck-in-life-use-the-force/

Another thing is to learn how to study - The GOAT ME method seems like a good starting point.

TL;DR: Get the right mindset, work hard, do it for yourself and it will pay off. Work beats talent in the long run.

Good Luck

Edit: I really appreciate the positive feedback and I'd advice you to read the comments since there are certain points I didn't take into account!

How to Start Anything and Get Unstuck in Life? Use the Force



Some of you are still stuck in life. You feel hopeless. You don't have the motivation to start anything. You have the same routine every single day. It's predictable. Life isn't exciting.

You feel lazy or unmotivated cause you're stuck. You have dreams, but you always fail to take action to make them come true. What will it take for you to finally get started?

I know what it took for me to get unstuck. It wasn't hope. It wasn't an inspirational quote. It wasn't a miracle. It's something that I possessed, but rarely used.

After years of getting no results, I realized one night I needed to do something about my life.

Like Obi-Wan Kenobi would say to me, *"Use the Force, Benny".*

Let's first talk about why you might feel stuck in life or not getting anything done.

Why do you feel stuck?

Feeling stuck doesn't mean your life sucks. You have many things to be thankful for in your life, but the part that's missing completely makes us forget what's good.

Everyone's life has the potential to be how they want it. As humans, we're the only species on earth that can change the course of our lives.

Yes, you are human! If you're not and you're reading this then that's pretty eerie. But since you're human that means you can change your life.

Feeling stuck means that you're not being fulfilled in a way that excites you. Where you are now is so far from where want to be. In that gap lies frustration and hopelessness.

I thought my life really sucked when I felt stuck, but when I looked around me at the time, I had lots to be thankful for. Simple things like good health, a house, a working car, food everyday and running hot water.

But the part that was missing from my life overshadowed all the good things I had.

If you feel stuck in life, I'm here to tell you there is hope. I've been there.

Your brain: Friend and Foe

Your brain can be your best friend or worst enemy. Let me explain.

A lot of things we do every day goes into auto pilot. We've done it so many times that our brain allows us to do it without much thinking. That can be great. It saves us from having to think about every single little thing we do.

If we had to, our minds would be exhausted by the end of the day.

Have you ever driven somewhere and when you got there you didn't remember driving there? You weren't drunk or on drugs (hope not at least). You were on autopilot.

When you wake up in the morning do you have a routine? Probably. You make coffee. You check Facebook. You check emails. That's your brain on auto pilot. Your brain is your friend in these situations.

The brain can be your enemy cause it's protects you too much. An important duty of the brain is to protect you from dangerous situations, but it also protects you like an overbearing parent.

Anytime you do something new, your mind will say, "Whoa….slow down there! Wait wait wait. Stop!!"

You stop. Then you start to think about that decision. Your inner voice starts to creep in. You then doubt yourself. You have second thoughts. Then you don't proceed any further.

You've stopped before you've even started.

You might be doing something new and exciting. Something you've dreamt of doing. It's not dangerous. It's not illegal. It's just new and that's why your brain stops you.

That is the problem.

Your brain wants to protect you. That's why it likes routines. It doesn't like change. Your brain knows what you're doing has been approved of many times. No warning signals go off.

But if we want to make our lives more exciting, get out of the same old boring routine, or achieve our goals, we need to break though that resistance.

Force Yourself to Do It

We need to break through what is trying to hold us back. In this case, there's only one way to break through the resistance.

You have to force yourself.

You just have to suck it up and do it. No complaining. No whining. Block everything else out of your mind.

Stop waiting for that one moment when the stars align. Stop waiting for when you feel like starting.

That push your looking for isn't in one famous quote that'll inspire you. Nor from another personal development book searching for that one sentence or paragraph that instantly turns you from unhappy to fulfilled in the blink of an eye.

Trust me it's not there because I wasted so many years searching for something that'd get me unstuck.

I kept waiting and waiting. Then I realized it was up to me to get started.

Remember being a kid? Our parents had to force us to do so many things.

Take a shower. Brush our teeth. Eat vegetables. Do our homework. Stop playing video games and go to sleep.

We didn't like it most of the time, but we agreed to.

If our parents never forced us, we wouldn't have done any of those things on our own. I know I wouldn't have!

Now that we're adults, who's going to force you to do what you need to do?

Nobody.

That's right. This push that you need isn't going to come from anyone else.

Will your parents force you to starting writing that book? Probably not. They can't punish you now by sending you to your room.

No one will force you to finally start pursing your passion. Only you can do that.

Who's going to force you to get off your butt and take some action in your life?

No one. No one's coming to save you. That's the truth.

It's not because they don't care, but cause they have a million of their own things to worry about.

Even if you paid a lot of money for a coach, they're not going to care if you don't take their advice and do something about it.

No one cares about your whining and your sob stories about why you can't. After awhile that whining gets annoying.

If you know something has to be done, but you're not doing it, then you're going to need to approach it differently.

Take your foot off the brakes and step on the gas!

The Three Second Rule



The Three Amigos

Pick up artists teach the three second rule. In a book called "The Game" By Neil Strauss (great book if you want to read about the world of pick up artists), he goes from average frustrated guy to a pick up guru. Along the way he learns what the best pick up artists do.

One important lesson for beginners is the three second rule.

It's vital for beginners. Why? Think about what the hardest thing is for a guy when meeting girls. It's not the small talk or how to get a phone number. The hardest thing is approach a woman and start talking!

The three second rule states that you must approach a woman within three seconds of noticing her. You might think that's way too fast. What will you say? What will you do? What will you say if she says this or that? So many things to think about!

The point is to stop that inner voice from talking to you. You don't have time to be nervous. You don't have time to second guess yourself. You don't have time for limiting beliefs.

It doesn't give you time to freak out. It also keeps you from freaking her out by staring at her like a stalker all night.

The only way to overcome your fear of approaching beautiful women is to not think too much about it and just do it.

How does the three second rule apply to you?

Do you ever get an impulse to do something, don't do it, and then regret it?

If you applied the three second rule, you wouldn't give your inner voice to talk you out of doing something you want to do.

Let's say you want more free time so you want to get up an hour early. We all know how hard it is to get up earlier. The moment the alarm goes off, we hit the snooze button and back to sleep. Just a few more minutes we think.

How long does it take for you to fall back asleep? Depending how early it is, it could be fast. Alarm rings again. We hit snooze again. Then all chances of waking up early goes away as we wake up our normal time.

What if you gave yourself the three second rule? You had three seconds to pull off the blankets and stand up and start moving.

You wouldn't have time to think "Should I snooze or should I get up early? I'm so tired so I need more sleep, but I really want to (your important task) before I have to go to work."

Getting right out of bed is so hard to do! It takes a huge amount of force. The bed is so comfortable and we feel so sleepy.

Don't think about how you feel. Of course you feel sleepy!

If you listen to how you feel versus what you want, you won't get it. How you feel will win every time. Feel tired? Sleep some more. Feel lazy? Procrastinate.

The force needed to get out of bed is the same force you need to do anything you want. It's not easy to do. It's a challenge, but if you're serious about getting up early or doing anything, it is what needs to be done.

If everything was so easy to get started, we'd all be well on our way to being perfectly happy.

When you want to do anything new in your life, you're going to have to get out of your comfort zone. In order to get out of that comfort zone, you can use the three second rule.

If you have an impulse to do something, if you don't take action in three seconds, you're going to hit the brakes.

That includes waking up the moment you hear your alarm.

Stop Waiting

There is probably a big goal you have right this moment. You're thinking about it, planning it, thinking about it, talking about it, and daydreaming about it. Guess what? You're not getting any younger.

There's not going to be a drill sergeant standing next to you while on your computer yelling "Get off Facebook! Close Twitter! Stop watching cute pictures of cats! Get to work!"

You have to force yourself to. You may not feel like doing it, but you know you need to.

I had to kick myself in the butt earlier this year, because I kept thinking what I wanted to achieve this year, but not doing anything about it! No one told me I needed to begin. I knew myself I needed to.

Finally I had to just start doing it. That's it. I stopped listening to my inner voice. I had to take action without any motivation. I had to use force to push that 2000 lb rock down the hill.

You have that power inside of you. It's time to use it.